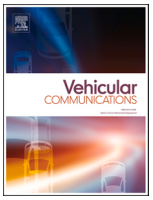
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Joint beamforming design for IRS-assisted ISAC in connected and automated vehicles ⋆

Yaping Cui [a,b,c,*], Jianjun Gu [a,b,c], Dapeng Wu [a,b,c], Peng He [a,b,c], Ruyan Wang [a,b,c], Gongxun Wang [a,b,c]

[a] School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, 400065, Chongqing, China
[b] Advanced Network and Intelligent Connection Technology Key Laboratory of Chongqing Education Commission of China, Chongqing, 400065, Chongqing, China
[c] Chongqing Key Laboratory of Ubiquitous Sensing and Networking, Chongqing, 400065, Chongqing, China

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ABSTRACT

Integrated sensing and communication (ISAC) provides an effective solution for supporting both high-rate data transmission and environmental perception in connected and automated vehicles (CAVs). However, in urban environments, buildings frequently hinder signal propagation between the base station (BS) and vehicles, thereby compromising link reliability and weakening the effectiveness of radar-based sensing. To mitigate these challenges, intelligent reflecting surface (IRS) technology has been proposed to enhance CAV operations by improving the signal propagation environment. This study focuses on the ISAC system implemented in IRS-assisted CAV scenarios. In this system, the IRS facilitates the simultaneous transmission of communication and radar signals. By jointly optimizing the active beamforming matrix at the BS and the passive beamforming matrix at the IRS, the sum rate achievable by vehicular users is maximized, while ensuring compliance with the BS's power budget, maintaining required radar detection quality, and adhering to the IRS phase configuration limits. In light of these considerations, an approach, namely Adaptive Optimization for Sensing and Forming (AOSF), is proposed, which employs the semidefinite relaxation (SDR) and successive convex approximation (SCA) algorithms to optimize the active beamforming matrix at the BS and employs the fractional programming algorithm to optimize the passive beamforming matrix at the IRS. The method decomposes the non-convex optimization problem into a series of manageable subproblems and employs an iterative solving approach to gradually refine the results and converge toward a satisfactory solution. Simulation results indicate that the proposed approach significantly improves achievable sum rate performance and reveals the trade-offs between communication and sensing.

1. Introduction

Future mobile communication networks are anticipated to play a pivotal role in addressing the extensive data demands of intelligent transportation systems (ITS) [1]. Connected and automated vehicles (CAVs), equipped with autonomous driving capabilities, can wirelessly communicate with other vehicles and infrastructure. Through real-time vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) data exchanges, these systems enhance road safety, traffic efficiency, and driving experiences [2]. In the ITS framework, CAVs are envisioned to transcend their current role as mere transportation tools, evolving into service platforms akin to mobile phones that offer passengers rapid and reliable data services. To achieve this vision, vehicles must possess two fundamental capabilities: sensing and communication [3]. Vehicles must be equipped with radar sensing to perceive their surrounding environment and gather critical information, such as the distance and speed of preceding vehicles, ensuring safety and enabling autonomous driving. Simultaneously, vehicles must communicate with other vehicles, passengers, and infrastructure via onboard transceivers to access network information, including ultra-high-resolution maps and real-time

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* Corresponding author at: School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, 400065, Chongqing, China.

E-mail addresses: cuiyp@cqupt.edu.cn (Y. Cui), s230131025@stu.cqupt.edu.cn (J. Gu), wudp@cqupt.edu.cn (D. Wu), hepeng@cqupt.edu.cn (P. He), wangry@cqupt.edu.cn (R. Wang), s210101129@stu.cqupt.edu.cn (G. Wang).

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updates, thereby facilitating navigation assistance and mitigating traffic congestion [4,5].

It is important to note that V2V and V2I represent specific communication paradigms that enable CAVs to exchange information with other entities in the transportation network. While V2V focuses on direct communication between vehicles to coordinate maneuvers and avoid collisions, V2I extends this connectivity to roadside units and infrastructure, providing access to broader traffic management systems and cloud services. In contrast, the concept of a CAV encompasses not only these communication functions but also advanced autonomous driving, environmental perception, and decision-making capabilities, integrating multiple sensing and computing technologies into a single intelligent vehicle platform. In contemporary onboard systems, however, communication and sensing functions are traditionally designed as standalone modules that operate in separate frequency bands and evolve independently. This design paradigm results in inefficient utilization of hardware resources and wireless spectrum [6].

Mobile communication networks must provide not only high-quality, ubiquitous wireless connectivity but also high-precision and robust sensing capabilities [7]. In recent years, the widespread adoption of key technologies such as millimeter wave (mmWave), terahertz (THz), and massive multiple-input multiple-output (MIMO) has facilitated the convergence of communication and radar frequency bands. This development paves the way for future mobile communication networks to simultaneously support communication and sensing tasks [8,9]. Integrated sensing and communication (ISAC) is an emerging technology that seeks to unify sensing and communication functionalities by leveraging common spectrum and hardware infrastructure. This method addresses the issue of spectral contention between radar and communication modules, thereby improving spectrum utilization and lowering hardware expenses [10]. Furthermore, the capacity for precise sensing opens new avenues for enhancing communication service quality. For instance, leveraging channel structure data obtained through sensing enables the rapid realignment and optimization of beamforming in response to dynamic channel fluctuations, which improves beamforming efficiency and enhances communication quality.

By sharing hardware and spectrum resources, ISAC enables CAVs to support high-speed communication and obstacle sensing simultaneously [11]. It is not surprising that significant research on ISAC-assisted CAVs has garnered considerable attention from the academic community [12]. The primary objective of CAVs is to enable a more convenient, real-time, intelligent connectivity system, which improves traffic safety and efficiency [13]. The development of vehicular networks requires the establishment of high-bandwidth V2I connections to base stations (BS) and, subsequently, to the internet. This will enable the implementation of a range of promising applications, including autonomous driving, traffic management, and vehicular platooning [14,15]. The aforementioned applications require rigorous specifications for enhanced throughput, reduced latency, and extensive connectivity [16]. However, the propagation environment in vehicular networks is complex, particularly in urban settings where buildings obstruct signals, which degrades the quality of vehicular communication links [17]. Thus, providing high-quality, ubiquitous wireless connectivity and improving the Quality of Service (QoS) for vehicular communication represents a significant challenge. To address this, future mobile communication networks aim to utilize IRS to establish controllable wireless channels, thus supporting both wireless communication and radar sensing.

The IRS has garnered significant attention from the academic community as an emerging technology with the potential to achieve high spectrum and energy efficiency, enhance channel capacity, and extend coverage [18]. The IRS system comprises multiple reflective elements that are passive and low-cost, significantly reducing hardware costs and energy consumption. Each reflective element can independently tune the phase shift of incident radio waves, enabling active control over the wireless propagation environment. This represents a paradigm shift in communication networks, transforming the channel from a passive medium to one that can be actively modified [19]. Thanks to their compact dimensions, IRSs can be flexibly deployed on building surfaces near vehicles, which helps lower implementation costs [20]. When LOS links are obstructed, the IRS can reconstruct virtual LOS paths by adjusting the reflection phase shifts. This ensures signals are accurately directed toward vehicular users, improving the SINR and supporting high-data-rate vehicular applications [21]. Additionally, the IRS can enhance the strength of desired target echo signals and suppress interference by altering the propagation environment. This improves the sensing performance in existing LOS links and provides additional reflective paths for ISAC systems. By offering these capabilities, the IRS enables radar to detect targets in shadowed areas not directly observable, thereby enhancing overall sensing performance [22].

In the context of IRS-assisted ISAC systems, two principal scenarios are commonly examined [23]. The first scenario focuses on using the IRS primarily to assist communication, while sensing relies on direct transmitter-to-target links. The second scenario fully exploits the IRS to enhance both communication and sensing by creating virtual LOS links, which improves radar detection performance especially in obstructed environments. In the first scenario, the IRS is used solely to facilitate communication, which enhances overall performance. Conversely, direct transmitter-to-target links are utilized for sensing purposes. To illustrate, the study in [24] examined an IRS-assisted ISAC framework, where a multi-antenna base station, with the support of an IRS, served multiple single-antenna users, while also performing target detection independently. In this configuration, the IRS is located near the users to enhance downlink multi-user communication. Since the IRS is located at a considerable distance from the sensing targets, the radar return signals reflected via the IRS are sufficiently weak to be neglected. Through the combined optimization of the BS's active beamforming and the IRS's passive beamforming, optimal system performance is attainable, considering constraints such as radar beam pattern similarity, IRS reflection coefficients, and the BS's power limitations. The second scenario, described in [25], examined an IRS-assisted MIMO ISAC system in which the IRS is solely responsible for downlink communication. This study proposed a BCD algorithm for the joint design of waveforms and passive beamforming, with the objective of simultaneously maximizing the radar's SINR and minimizing multi-user interference in communication. Furthermore, the study in [26] considered an IRS-assisted dual-functional radar communications (DFRC) system, where the IRS supports downlink communication but also introduces interference to target sensing. The objective was to minimize the power radiated by the BS while satisfying the minimum SINR thresholds required by communication and radar detection functions. This was achieved through the joint design of active beamforming at the BS and the IRS's passive beamforming matrix.

The second scenario fully utilizes the IRS's potential, whereby the IRS not only enhances communication performance but also establishes virtual LOS links between the BS and sensing targets, aiming to improve radar sensing performance. To illustrate, the study in [27] investigated an ISAC framework supported by an IRS, which facilitated wireless communication linking the base station with single-antenna users, while simultaneously creating virtual line-of-sight connections to detect targets in obstructed regions. The optimization problem entails simultaneous adjustment of the beamforming matrices at both the BS and IRS, aiming to maximize the minimum directional gain of the IRS's beam pattern toward the targeted sensing direction. This is accomplished subject to limitations on the BS's peak transmission power, the minimum SNR required by communication users, and the phase configuration constraints of the IRS. In [28], the study focused on an ISAC setup enhanced by an IRS, in which the BS carried out multi-target detection alongside multi-user communication. The coordinated design of the BS's transmit beamforming and the IRS's reflective beamforming matrices seeks to optimize the weighted sum of the target detection SNR, while meeting communication SINR targets, adhering to the BS's total power constraints, and satisfying the IRS's unit-modulus phase shift requirements. Similarly, Han et al. [29] addressed sensing security within ISAC systems by uti-

lizing ambiguity function engineering. This framework ensures robust target detection for legitimate systems while impairing unauthorized passive sensing by introducing artificial ghost targets into the eavesdropper's range profile. In summary, most current IRS-enabled ISAC schemes concentrate on developing and solving targeted optimization problems that prioritize maximizing either communication or sensing performance, while guaranteeing baseline requirements for the other. This balance is attained by fine-tuning the constraint thresholds related to each functionality.

Different from generic IRS-assisted ISAC beamforming frameworks, this work focuses on obstructed and time-varying urban CAV environments. In these environments, urban blockage makes the BS-IRS-VUE cascaded channel an essential controllable propagation path for reconstructing reliable V2I links, while vehicle mobility results in block-dependent communication channels. Accordingly, the active and passive beamforming variables are optimized for each Doppler-compensated quasi-static channel realization. Moreover, the BS simultaneously supports V2I transmission and radar sensing of vehicle targets. These characteristics determine the channel model, temporal operation, and communication-sensing constraints of the considered joint beamforming problem.

In light of the preceding analysis, this paper investigates the potential of an IRS-assisted ISAC system for CAVs. In this configuration, the IRS aids the BS in jointly transmitting communication and radar signals, enabling the simultaneous provision of communication services to users and the tracking of targets. The initial objective is to maximize the achievable sum rate of VUEs through the coordinated design of the BS's active beamforming and the IRS's passive beamforming matrix. This is accomplished while adhering to limitations on the BS's peak transmission power, the worst-case radar SINR, and the IRS's unit-modulus phase shift constraints. Subsequently, the AOSF algorithm is proposed to iteratively optimize both the active beamforming at the BS and the passive beamforming at the IRS. Specifically, SDR and SCA are used to solve the BS active beamforming subproblem for a given IRS passive beamforming matrix, while FP is employed to update the IRS passive beamforming matrix for a given BS active beamforming matrix. The main contributions of this paper are as follows:

- This paper formulates a CAV-oriented IRS-assisted ISAC beamforming problem for obstructed urban V2I scenarios. By jointly designing the BS active beamforming matrix W and the IRS passive beamforming matrix $\mathbf{\Theta}$, the achievable sum rate of VUEs is maximized while satisfying the BS transmit power budget, the radar SINR requirement, and the IRS unit-modulus phase-shift constraint. The considered formulation explicitly reflects urban blockage of direct BS-VUE links, IRS-assisted virtual LOS reconstruction for V2I communication, vehicle target sensing, and block-based vehicular channel variation.
- To address the original optimization problem, the AOSF algorithm is proposed, which decomposes it into two subproblems: one associated with the BS's active beamforming and the other concerning the IRS's passive beamforming. Specifically, with a given IRS passive beamforming matrix $\mathbf{\Theta}$, the BS active beamforming matrix W is optimized using SDR and SCA algorithms. Subsequently, with W held constant, the FP algorithm is utilized to optimize the IRS passive beamforming matrix $\mathbf{\Theta}$.
- Simulation results highlight the advantages of integrating an IRS into an ISAC system for CAVs. The proposed algorithm delivers enhanced achievable sum-rate performance compared to IRS-free schemes, emphasizing the balance between communication efficiency and sensing accuracy.

The remainder of the paper is organized as follows. Section 2 introduces the system model, while Section 3 outlines the objective function. Section 4 presents the alternating optimization algorithm to address the non-convex joint beamforming optimization problem. Section 5 presents the simulation results, and finally, Section 6 concludes the paper.

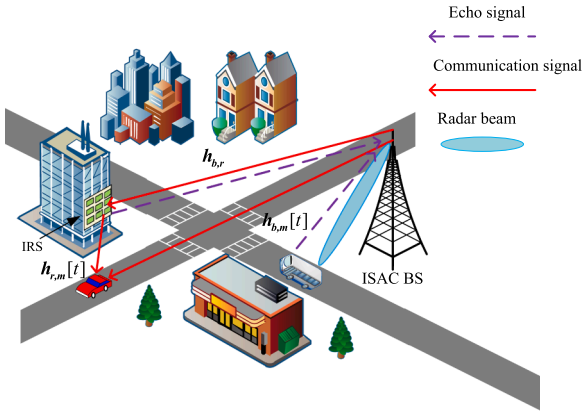


Fig. 1. System model.

2. System model

As illustrated in Fig. 1, consider an IRS-assisted downlink Multiple-Input-Single-Output (MISO) ISAC system, comprising a BS equipped with N_T transmit antennas and N_R receive antennas, supported by an IRS with N passive reflecting elements. The system simultaneously transmits statistically independent radar and communication signals to L radar-sensing vehicle targets and M Vehicular User Equipment (VUEs). The set of M single-antenna vehicles communicating through the V2I link is represented by $\mathcal{M} = \{1, \ldots, m, \ldots, M\}$. The antennas at the BS are arranged in a uniform linear array (ULA) configuration. This configuration enables the simultaneous transmission to both VUEs and radar-sensing vehicle targets using N_T transmit antennas, and the reception of echo signals reflected from radar targets using N_R receive antennas. The BS's transmit and receive antennas operate simultaneously with perfect self-interference cancellation, enabled by advanced full-duplex and self-interference cancellation technologies [30].

2.1. Transmission signal

The DFRC base station employs transmit beamforming to simultaneously deliver communication data and dedicated radar sensing signals, thus serving both users and sensing targets. Therefore, the transmitted signal at the base station can be expressed as follows [31]:

$$x = W_c s_c + W_r s_r = W s, \tag{1}$$

where $W_c \in \mathbb{C}^{N_T \times M}$ and $W_r \in \mathbb{C}^{N_T \times N_T}$ denote the communication beamforming matrix and radar beamforming matrix, respectively. The communication symbols are represented by $s_c \in \mathbb{C}^{M \times 1}$, which satisfy $s_c \sim \mathcal{CN}(0, I_M)$. The radar detection signals, $s_r \in \mathbb{C}^{N_T \times 1}$, are generated using pseudo-random coding and satisfy $\mathbb{E}\{s_r\} = 0_{N_T \times 1}$. The radar and communication signals are considered statistically independent, i.e., $\mathbb{E}\{s_c s_r^H\} = 0_{M \times N_T}$. The combined beamforming matrix is represented as $W \triangleq [W_c, W_r] \in \mathbb{C}^{N_T \times (M+N_T)}$, while the transmitted signal vector is expressed as $s \triangleq \left[s_c^T, s_r^T\right]^T \in \mathbb{C}^{(M+N_T) \times 1}$.

2.2. Communication model

A transmission frame is composed of T blocks, with the set denoted as $\mathcal{T} = \{1, \ldots, t, \ldots, T\}$. The IRS is arranged as a Uniform Planar Array (UPA) composed of N_h passive reflecting elements horizontally and N_v vertically, yielding a total of $N = N_h \times N_v$ elements. The IRS is controlled by the IRS controller through the BS, enabling the controller to adjust the phase of each array element to reflect the incident signal in the desired direction. The IRS reflection behavior is characterized by a diagonal matrix $\Theta = \mathrm{diag}(v_1, \ldots, v_n, \ldots, v_N) \in \mathbb{C}^{N \times N}$, where $v_n = \mu_n e^{j\vartheta_n}$. Here, $\vartheta_n \in [0, 2\pi)$ and $\mu_n \in [0, 1]$ denote the phase shift and amplitude

response associated with the nth reflecting unit, respectively. For simplicity, the amplitude reflection coefficients of all reflecting elements are set to $\mu_n = 1$ in order to maximize signal reflection.

In the downlink transmission, let $h_{b,r} \in \mathbb{C}^{N\times N_T}$, $h_{r,k}[t] \in \mathbb{C}^{N\times 1}$, and $h_{b,k}[t] \in \mathbb{C}^{N_T\times 1}$ denote the BS-IRS, the IRS-VUE k and BS-VUE k channels in the tth transmission slot, modeled as follows:

$$h_{b,r} = \sqrt{L_{b,r}}\left(\sqrt{\frac{\zeta}{1+\zeta}}\bar{h}_{b,r}^{\mathrm{LOS}} + \sqrt{\frac{1}{1+\zeta}}\tilde{h}_{b,r}^{\mathrm{NLOS}}\right), \tag{2}$$

$$h_{r,m}[t] = \sqrt{L_{r,m}}\left(\sqrt{\frac{\zeta}{1+\zeta}}\bar{h}_{r,m}^{\mathrm{LOS}}[t] + \sqrt{\frac{1}{1+\zeta}}\tilde{h}_{r,m}^{\mathrm{NLOS}}[t]\right), \tag{3}$$

$$h_{b,m}[t] = \sqrt{L_{b,m}}\tilde{h}_{b,m}[t], \tag{4}$$

where ζ is the corresponding Rician factor associated with small-scale fading, $L_{b,r}$, $L_{r,k}$ and $L_{b,k}$ represent the corresponding large-scale path loss, respectively, which can be calculated as:

$$L_{b,r} = \kappa_0\left(\frac{d_{b,r}}{D_0}\right)^{-\alpha_{b,r}}, \tag{5}$$

$$L_{r,m} = \kappa_0\left(\frac{d_{r,m}}{D_0}\right)^{-\alpha_{r,m}}, \tag{6}$$

$$L_{b,m} = \kappa_0\left(\frac{d_{b,m}}{D_0}\right)^{-\alpha_{b,m}}, \tag{7}$$

where κ_0 is the path loss coefficient at $D_0 = 1$ m, $\alpha_{b,r}$, $\alpha_{r,k}$ and $\alpha_{b,k}$ are the corresponding path loss exponents, the distances $d_{b,r}$, $d_{r,k}$ and $d_{b,k}$ represent the respective distances of the BS-IRS link, IRS-VUE k link and BS-VUE k link. $h_{b,r}^{\mathrm{NLOS}}$ and $h_{r,k}^{\mathrm{NLOS}}[t]$ represent the NLOS components, while $\tilde{h}_{b,k}[t]$ represents the Rayleigh fading component, with each element of them choosing from $CN(0,1)$. $h_{b,r}^{\mathrm{LOS}}$ and $h_{r,k}^{\mathrm{LOS}}[t]$ represent the LOS components, which can be given by:

$$\bar{h}_{b,r}^{\mathrm{LOS}} = a(\theta_R,\varphi_R)a^{\mathrm{H}}(\theta_B,\varphi_B), \tag{8}$$

$$\bar{h}_{r,m}^{\mathrm{LOS}}[t] = a(\theta_m,\varphi_m), \tag{9}$$

where $a_P\left(\vartheta_{b,r}^{\mathrm{AoA}},\phi_{b,r}^{\mathrm{AoA}}\right)$ and $a_P\left(\vartheta_{r,k}^{\mathrm{AoD}},\phi_{r,k}^{\mathrm{AoD}}\right)$ denote the uniform planar array (UPA) steering vectors at the IRS and the relay, respectively, while $a_L\left(\vartheta_{b,r}^{\mathrm{AoD}},\phi_{b,r}^{\mathrm{AoD}}\right)$ represents the uniform linear array (ULA) steering vector at the BS. These vectors can be formulated as in [31]:

$$a(\theta_B,\varphi_B) = \left(1, e^{j\frac{2\pi d}{\lambda}\sin\theta_B\sin\varphi_B}, \ldots, e^{j\frac{2\pi d}{\lambda}(N_T-1)\sin\theta_B\sin\varphi_B}\right)^{\mathrm{T}}, \tag{10}$$

$$\begin{aligned} a(\theta_X,\varphi_X) =& \left(1, e^{j\frac{2\pi d}{\lambda}\sin\theta_X\cos\varphi_X}, \ldots, e^{j\frac{2\pi d}{\lambda}(N_h-1)\sin\theta_X\cos\varphi_X}\right)^{\mathrm{T}} \\ &\otimes \left(1, e^{j\frac{2\pi d}{\lambda}\sin\theta_X\sin\varphi_X}, \ldots, e^{j\frac{2\pi d}{\lambda}(N_v-1)\sin\theta_X\sin\varphi_X}\right)^{\mathrm{T}}, \end{aligned} \tag{11}$$

where d represents the antenna/element spacing and λ denotes the carrier wavelength. For ease of analysis, the spacing is set to $d = \lambda/2$. The angles $\vartheta_{b,r}^{\mathrm{AoD}}$ and $\phi_{b,r}^{\mathrm{AoD}}$ correspond to the elevation and azimuth angles of departure (AODs) at the BS toward the IRS, while $\vartheta_{b,r}^{\mathrm{AoA}}$ and $\phi_{b,r}^{\mathrm{AoA}}$ denote the respective angles of arrival (AOAs) at the IRS from the BS. Similarly, $\vartheta_{r,k}^{\mathrm{AoD}}$ and $\phi_{r,k}^{\mathrm{AoD}}$ specify the elevation and azimuth AODs at the IRS directed toward VUE k. Additionally, $\otimes$ denotes the Kronecker product. The validity of the steering vector models in (10) and (11) relies on the far-field planar wavefront assumption. This assumption holds when the distance between the transceiver and the IRS exceeds the Fraunhofer distance, defined as $d_f = \frac{2D^2}{\lambda}$, where D denotes the maximum aperture dimension of the IRS and λ is the carrier wavelength. Based on the simulation configuration detailed in Section 5 (with $N = 20$ reflecting elements), the calculated Fraunhofer distance is significantly smaller than the typical operating distances (e.g., tens of meters) in the considered CAV scenarios. Consequently, the far-field condition is satisfied, justifying the use of planar wavefront models.

By leveraging accurate channel estimation strategies [32], it is presumed that the DFRC BS is equipped with comprehensive knowledge of the CSI related to the aforementioned channels. Given that the vehicle's position predominantly determines the large-scale fading and deterministic LOS components, and the displacement within a single block is minimal compared to the BS distance, it is reasonable to treat these components as time-invariant during each block. To capture the mobility-induced challenges in CAV environments, we model the time-varying nature of the channel through the framework of transmission blocks. Specifically, while vehicular movement causes rapid channel variations, the duration of each block t is designed to be within the channel coherence time. Under this block-fading assumption, the small-scale fading and Doppler effects are treated as quasi-static within a single block, assuming that Doppler frequency shifts are pre-processed and compensated for at the receiver. Consequently, the joint beamforming design focuses on optimizing the spatial degrees of freedom for each instantaneous channel realization, providing a foundation for adapting to the continuous temporal dynamics.

In practical IRS-assisted CAV systems, the CSI used for joint beamforming can be obtained from channel estimation modules. For passive IRS-assisted links, acquiring the BS-IRS-VUE cascaded CSI may introduce additional pilot overhead, especially when the number of IRS reflecting elements is large. This overhead can be reduced through pilot training with IRS element grouping, location-aided channel estimation, beam training, or predicted CSI. In this work, the estimated CSI is used as the input to the proposed beamforming framework, while the effects of residual CSI and Doppler estimation errors are further discussed in Section 5.

Accordingly, the mathematical expression for the signal observed by the mth VUE is given by:

$$y_m[t] = \left(h_{r,m}^H[t]\Theta h_{b,r} + h_{b,m}^H[t]\right)x[t] + n_m[t], \tag{12}$$

where $n_0[t] \sim CN(0,\sigma^2)$ represents the additive white Gaussian noise.

Consequently, the mth VUE experiences an SINR that is given by:

$$\mathrm{SINR}_m[t] = \frac{\left|h_m^H[t]w_m\right|^2}{\sum\limits_{\substack{i=1\\ i\neq m}}^{M+N_T}\left|h_m^H[t]w_i\right|^2 + \sigma_m^2}, \tag{13}$$

here, $\boldsymbol{w}_i \in \mathbb{C}^{N_T\times 1}$ denotes the ith beamforming vector in matrix $\boldsymbol{W}$, and σ^2 represents the corresponding noise power.

In this study, we investigate the impact of the SINR of the V2I links on the sum of V2I capacity, which remains independent of the bandwidth. To facilitate the analysis, the bandwidth is fixed at 1. Accordingly, the channel capacity corresponding to the mth VUE is given by:

$$R = \sum_{m=1}^{M}\log_2\left(1+\mathrm{SINR}_m[t]\right). \tag{14}$$

In subsequent analyses, we focus on the performance metrics of each block. For simplicity, we omit the index "t" representing the tth block.

2.3. Radar model

The transmitted signal at the BS serves two distinct functions: delivering information to communication vehicles and detecting targets in the surrounding environment. From a radar sensing perspective, it is crucial to concentrate power in the direction of the target while minimizing signal power in unrelated directions to enhance target detection efficiency. The target detection performance depends directly on the SINR of the echo signal received by the BS. Thus, the SINR of the BS's received echo serves as a key sensing performance metric.

Considering that the spatial extent of the target is limited, the incident signal is reflected solely by discrete point targets. The signal corresponding to the lth detected vehicle target, located at angle θ_l, can be modeled by the transmit steering vector $\mathbf{a}(\theta_l)$, where $\mathbf{a}(\theta)$ denotes the steering vector oriented towards θ, d represents the antenna spacing, and λ is the carrier wavelength. The echo signal arriving at the ISAC BS originates from four principal propagation paths: the direct BS-target-BS link, the BS-IRS-BS reflection, and two triple-hop channels — BS-IRS-target-BS and BS-target-IRS-BS. Nevertheless, both theoretical insights and numerical results demonstrate that the triple-hop paths suffer severe attenuation, rendering their contributions negligible for system performance enhancement [33]. Specifically, the triple-hop path involves an additional propagation link compared to the direct two-hop path. By applying the simulation parameters from Table 1 (with a path loss exponent of 2.8 and a reference loss of -30 dB), a typical extra link distance of 50 m introduces an additional attenuation of approximately 77.5 dB. Given that the triple-hop signal strength is nearly 8 orders of magnitude weaker than the direct echo, its impact falls well below the noise floor and can be safely disregarded. Consequently, the analysis focuses exclusively on the direct BS-target-BS and BS-IRS-BS echo components. The received echo signal at the BS is then given by:

$$y_r = \underbrace{\sum_{l=1}^{L} \beta_l a_r(\theta_l) a_t^H(\theta_l) x}_{\text{reflected by targets}} + \underbrace{h_{r,b}\Theta h_{b,r} x}_{\text{reflected by the IRS}} + n_r, \tag{15}$$

where β_l represents the reflection coefficient associated with the lth target, which is proportional to its Radar Cross Section (RCS). Moreover, the ISAC base station (ISAC-BS) is assumed to have knowledge of the target's angular position θ_l and the magnitude of the reflection coefficient a. The target angular information used for radar beamforming can be obtained through sensing estimation methods such as the Generalized Likelihood Ratio Test (GLRT) and Capon beamforming techniques [34]. In practical CAV scenarios, angle estimation errors may occur due to vehicle mobility, clutter, and multipath propagation. Such errors may affect sensing beam alignment and can be alleviated by target tracking and robust sensing beamforming methods. Let h_{BS-IRS} denote the complex equivalent baseband channel linking the BS and IRS during uplink transmission, and let n_{BS} be the additive white Gaussian noise at the BS. Similar to the transmit steering vector $\boldsymbol{a}(\theta)$, $\boldsymbol{b}(\theta)$ defines the receive steering vector oriented towards angle θ. Since the echo signal reflected via the BS-IRS-BS path carries no target-specific information, it is treated as interference in the sensing process. Within the ISAC-BS framework, the communication signal plays a dual role by supporting both data transmission and target tracking. Given that the communication signal is known at the BS, it does not cause interference to the tracking operation. Because the IRS-reflected signal described in (15) contains no target-related content, the radar's signal-to-interference-plus-noise ratio (SINR) is formulated as in [35]:

$$\mathrm{SINR}_r = \mathrm{tr}\left(AGA^H\left(DGD^H + \sigma_r^2 I_{N_r}\right)^{-1}\right), \tag{16}$$

where $\mathrm{A} = \sum_{l=1}^{L} \beta_l a_r(\theta_l) a_t^H(\theta_l)$, $\mathrm{G} = \sum_{m=1}^{M+N_T} \mathrm{w}_m \mathrm{w}_m^H$, $\mathrm{D} = h_{r,b}\Theta h_{b,r}$. The SINR of the echo signal determines the probability of sensing the target. Therefore, the SINR of the echo signal should meet a high threshold.

3. Problem formulation

In this work, we aim to jointly optimize the transmit beamforming matrix W at the base station and the reflection coefficient matrix Θ at the IRS. The optimization is performed under the constraints of the BS's total transmit power budget, the radar's minimum SINR requirement, and the IRS phase shift limitations. The primary objective is to maximize the aggregate communication rate of all vehicular users. Accordingly, the problem can be mathematically expressed as:

$$P(1): \max_{W,\Theta} R = \sum_{m\in M} \log_2\left(1 + \mathrm{SINR}_m\right), \tag{17a}$$

$$\text{s.t.} \quad \mathrm{tr}\left(AGA^H\left(DGD^H + \sigma_r^2 I_{N_r}\right)^{-1}\right) \geq \gamma_{r,\mathrm{th}}, \tag{17b}$$

$$\sum_{m=1}^{M+N_T} \left\|w_m\right\|^2 \leq P_{\max}, \tag{17c}$$

$$\left|v_n\right| = 1, \quad \forall n = 1, 2, \ldots, N. \tag{17d}$$

where $P_{\max}$ denotes the maximum allowable transmit power at the base station. Constraint (17b) guarantees that the signal-to-interference-plus-noise ratio (SINR) of the echo received at the BS surpasses the required minimum threshold $\gamma_{r,th}$, ensuring adequate beamforming gain for target detection. Constraint (17c) limits the total transmit power at the BS, while constraint (17d) enforces a unit-modulus condition on each IRS reflection coefficient. Due to the strong coupling between optimization variables in the objective function, problem $P(1)$ is inherently non-convex. Consequently, there is generally no straightforward approach to obtain a globally optimal solution for such non-convex problems.

4. Alternating optimization algorithm

Solving problem $P(1)$ optimally is difficult due to the interdependence between the transmit beamforming matrix W at the BS and the IRS reflection coefficients Θ. Therefore, this section approaches the problem by splitting it into two subproblems, referred to as $P(2)$ and $P(3)$, respectively. Subsequently, the Adaptive Optimization for Sensing and Forming (AOSF) algorithm is applied to iteratively and jointly optimize W and Θ. Accordingly, the AOSF framework combines SDR, SCA, and FP to address the coupled active and passive beamforming subproblems arising from the CAV-oriented channel model and communication-sensing constraints.

4.1. Semidefinite relaxation for BS beamforming

With the IRS phase shifts Θ fixed, the optimization problem for determining the BS transmit beamforming matrix W can be formulated as follows:

$$P(2): \max_{W} \sum_{m\in M} \log_2\left(1 + \mathrm{SINR}_m\right), \tag{18a}$$

$$\text{s.t.} \quad \mathrm{tr}\left(AGA^H\left(DGD^H + \sigma_r^2 \mathbf{I}_{N_r}\right)^{-1}\right) \geq \gamma_{r,th}, \tag{18b}$$

$$\sum_{m=1}^{M+N_T} \|w_m\|^2 \leq P_{\max}. \tag{18c}$$

To facilitate analysis, we rewrite the objective function in (18a) into a more manageable form. Define $H_m = h_m h_m^H$ and $W_i = w_i w_i^H$. Accordingly, the achievable rate of the mth communication VUE in the objective can be represented as:

$$\log_2\left(1 + \mathrm{SINR}_m\right) = \log_2\left(\frac{\sum_{i=1}^{M+N_T} \mathrm{tr}\left(H_m W_i\right) + \sigma_m^2}{\sum_{\substack{i=1 \\ i\neq m}}^{M+N_T} \mathrm{tr}\left(H_m W_i\right) + \sigma_m^2}\right). \tag{19}$$

Introducing exponential auxiliary variables [36], we set them as follows:

$$e^{\delta_m} = \sum_{i=1}^{M+N_T} \mathrm{tr}\left(H_m W_i\right) + \sigma_m^2, \tag{20}$$

$$e^{\tau_m} = \sum_{\substack{i=1 \\ i\neq m}}^{M+N_T} \mathrm{tr}\left(H_m W_i\right) + \sigma_m^2, \tag{21}$$

Based on (20) and (21), removing the constant terms, the objective function (18a) can be transformed into:

$$\max_{\{\delta_m,\tau_m\}_{m=1}^{M}} \sum_{m=1}^{M} (\delta_m - \tau_m). \tag{22}$$

Eq. (22) is a convex function; the constraints on δ_m and τ_m can be expressed as:

$$e^{\delta_m} \le \sum_{i=1}^{M+N_T} \mathrm{tr}\left(H_m W_i\right) + \sigma_m^2, \tag{23}$$

$$e^{\tau_m} \ge \sum_{\substack{i=1 \\ i\neq m}}^{M+N_T} \mathrm{tr}\left(H_m W_i\right) + \sigma_m^2. \tag{24}$$

Eqs. (23) and (24) hold as equalities at the optimal point, which can be verified through the monotonicity of the objective function. However, after transforming (23) and (24), (24) remains non-convex in form. To address this non-convexity, the SCA method is employed, utilizing the first-order Taylor expansion [36]. Consequently, (24) can then be expressed as:

$$\sum_{\substack{i=1 \\ i\neq m}}^{M+N_T} \mathrm{tr}\left(H_m W_i\right) + \sigma_m^2 \le e^{\tau_m^k}\left(\tau_m - \tau_m^k + 1\right), \tag{25}$$

where τ_m^k is the optimal value at the kth iteration.

At this juncture, it's apparent that constraint (18b) is non-convex. We can derive a lower bound for constraint (18b) based on an identity, which can be expressed as [26]:

$$\mathrm{tr}\left(AGA^H\left(DGD^H + \sigma_r^2 \mathbf{I}_{N_r}\right)^{-1}\right) \ge \frac{\mathrm{tr}\left(AGA^H\right)}{\mathrm{tr}\left(DGD^H + \sigma_r^2 \mathbf{I}_{N_r}\right)}. \tag{26}$$

Constraint (18b) can be reformulated into a more manageable approximation as follows:

$$\mathrm{tr}\left(A\left(\sum_{i=1}^{M+N_T} W_i\right)A^H\right) \ge \gamma_{r,th}\left(\mathrm{tr}\left(D\left(\sum_{i=1}^{M+N_T} W_i\right)D^H\right) + \sigma_r^2 N_r\right). \tag{27}$$

At this stage, the optimization problem for determining the beamforming matrix W at the BS can be reformulated as:

$$P(2.1): \max_{\{\delta_m,\tau_m\}_{m=1}^{M},\{W_i\}_{i=1}^{M+N_T}} \sum_{m=1}^{M}(\delta_m - \tau_m), \tag{28a}$$

$$\text{s.t.} \quad \sum_{i=1}^{M+N_T} \mathrm{tr}(W_i) \le P_{\max}, \tag{28b}$$

$$W_i \ge 0, \tag{28c}$$

$$\mathrm{rank}(W_i) = 1, \tag{28d}$$

$$(23), (25), (27). \tag{28e}$$

Due to the rank-1 constraint in (28d), problem $P(2.1)$ becomes non-convex and thus cannot be directly solved using standard convex optimization techniques. To tackle this issue, the Semidefinite Relaxation (SDR) method can be employed. The key idea of SDR is to reformulate the problem by lifting the original vector variable into a higher-dimensional matrix variable, which allows the quadratic forms to be expressed linearly in terms of this matrix. Specifically, the rank-1 constraint on the matrix variable is relaxed to a positive semidefinite constraint, thereby converting the original non-convex problem into a convex SDP problem.

Following this relaxation, it is clear that both the objective function and constraints of problem $P(2.1)$ become convex with respect to the newly introduced matrix variable. This convexity allows for efficient resolution using optimization tools like CVX. Nevertheless, as the rank constraint is omitted, the solution obtained from the relaxed problem may not necessarily be of rank one. In such cases, the Gaussian randomization method can be applied to construct a feasible approximate solution that satisfies the original rank one requirement [37]. This approach is widely used in wireless communications and signal processing to handle non-convex quadratic problems that involve beamforming or covariance matrix design.

4.2. Fractional programming for IRS beamforming

After obtaining the BS beamforming matrix W, the optimization problem for solving the IRS phase shifts Θ can be expressed as:

$$P(3): \max_{\Theta} \sum_{m\in M} \log_2\left(1 + \mathrm{SINR}_m\right), \tag{29a}$$

$$\text{s.t.} \quad \mathrm{tr}\left(AGA^H\left(DGD^H + \sigma_r^2 \mathbf{I}_{N_r}\right)^{-1}\right) \ge \gamma_{r,th}, \tag{29b}$$

$$|v_n| = 1, \quad \forall n = 1, 2, \ldots, N. \tag{29c}$$

To handle the complex objective function in problem $P(3)$, we utilize the FP method to transform it into a more easily solvable polynomial form. The FP method decomposes the fractional objective into simpler subproblems by introducing auxiliary variables, enabling iterative optimization with guaranteed convergence. By converting ratios into equivalent polynomial forms, FP effectively tackles the non-convexity inherent in SINR-based optimization problems. Utilizing the Lagrangian dual transform technique from [38], an auxiliary variable $b = [b_1, \ldots, b_M]$ is introduced to separate the SINR_m term from the logarithmic function. This reformulation converts the objective function in problem $P(3)$ into the following equivalent form:

$$\max_{\Theta,b} f(\Theta, b) = \sum_{m=1}^{M} \log_2(1 + b_m) - \sum_{m=1}^{M} b_m + \sum_{m=1}^{M} \frac{(1 + b_m)\left|h_m^H w_m\right|^2}{\sum_{j=1}^{M+N_T}\left|h_m^H w_j\right|^2 + \sigma_m^2}. \tag{30}$$

With Θ held constant, computing the first-order partial derivative of the above expression with respect to b_m and equating it to zero results in the optimal value of b_m, which is given by:

$$b_m^{\mathrm{opt}} = \frac{\left|h_m^H \mathrm{w}_m\right|^2}{\sum_{i=1, i\neq m}^{M+N_T}\left|h_m^H \mathrm{w}_i\right|^2 + \sigma_m^2}, \forall m. \tag{31}$$

When b is fixed, (30) can be simplified to:

$$\max_{\Theta} \sum_{m=1}^{M} \frac{(1 + b_m)\left|h_m^H \mathrm{w}_m\right|^2}{\sum_{j=1}^{M+N_T}\left|h_m^H \mathrm{w}_j\right|^2 + \sigma_m^2}. \tag{32}$$

Nevertheless, the fractional structure in (32) complicates direct optimization. By introducing an auxiliary variable c and applying the quadratic transformation technique, the problem can be reformulated as in [38].

$$2\sqrt{1 + b_m}\,\mathrm{Re}\left\{c_m^{\mathrm{opt}} h_m^H \mathrm{w}_m\right\} - |c_m|^2\left(\sum_{j=1}^{M+N_T}\left|h_m^H \mathrm{w}_j\right|^2 + \sigma_m^2\right), \tag{33}$$

where The optimal value of the auxiliary variable c_m can be expressed as:

$$c_m^{\mathrm{opt}} = \frac{\sqrt{1 + b_m} h_m^H \mathrm{w}_m}{\sum_{j=1}^{M+N_T}\left|h_m^H \mathrm{w}_j\right|^2 + \sigma_m^2}. \tag{34}$$

Based on (30) and (33), after obtaining the auxiliary variables b_m and c_m, the objective function with respect to Θ in problem P(3) can be rewritten as:

$$\max_{\Theta} \sum_{m=1}^{M} \left(2\sqrt{1+b_m}\,\mathrm{Re}\left\{c_m^{\mathrm{opt}} h_m^H \mathrm{w}_m\right\} - |c_m|^2 \sum_{j=1}^{M+N_T} \left|h_m^H \mathrm{w}_j\right|^2\right). \tag{35}$$

Then, utilizing the equation $h_{r,m}^H \Theta h_{b,r} \mathrm{w}_j = h_{r,m}^H \mathrm{diag}(h_{b,r}\mathrm{w}_j)v$, problem $P(3)$ can be reformulated as:

$$P(3.1): \min_{v} v^H Z v - 2\,\mathrm{Re}\left\{v^H z\right\} - r, \tag{36a}$$

$$s.t. \quad \mathrm{tr}\left(AGA^H\left(DGD^H + \sigma_r^2 \mathbf{I}_{N_r}\right)^{-1}\right) \geq \gamma_{r,th}, \tag{36b}$$

$$|v_n| = 1, \quad \forall n = 1, 2, \ldots, N, \tag{36c}$$

where

$$Z \triangleq \sum_{m=1}^{M} |c_m|^2 \sum_{j=1}^{M+N_T} \mathrm{diag}\left(\mathrm{w}_j^H h_{b,r}^H\right) h_{r,m} \times h_{r,m}^H \mathrm{diag}\left(h_{b,r}\mathrm{w}_j\right), \tag{37}$$

$$z \triangleq \sum_{m=1}^{M} \sqrt{1+b_m} c_m \mathrm{diag}\left(\mathrm{w}_m^H h_{b,r}^H\right) h_{r,m} - \sum_{m=1}^{M} |c_m|^2 \sum_{j=1}^{M+N_T} \mathrm{diag}\left(\mathrm{w}_j^H h_{b,r}^H\right) h_{r,m} h_{b,m}^H \mathrm{w}_j^H, \tag{38}$$

$$r \triangleq \mathrm{Re}\left\{\sum_{m=1}^{M} 2\sqrt{1+b_m} c_m^{\mathrm{opt}} h_{b,m}^H \mathrm{w}_m - |c_m|^2 \times \sum_{j=1}^{M+N_T} \left|h_{b,m}^H \mathrm{w}_j\right|^2\right\}. \tag{39}$$

Then, expanding the $\mathrm{tr}\left(DGD^H\right)$ in the right-hand side of (27), it can be expressed as:

$$\mathrm{tr}\left(DGD^H\right) = \mathrm{tr}\left(\Theta^H h_{r,b}^H h_{r,b} \Theta h_{b,r} G h_{b,r}^H\right) = v^T\left(\left(h_{r,b}^H h_{r,b}\right) \odot \left(h_{b,r} G h_{b,r}^H\right)^T\right) v^*, \tag{40}$$

where v^* is the conjugate transpose of v, it can be observed that both $h_{r,b}^H h_{r,b}$ and $h_{b,r}\mathrm{G}h_{b,r}^H$ are positive semi-definite matrices. Therefore, the Hadamard product of $h_{r,b}^H h_{r,b}$ and $h_{b,r}Gh_{b,r}^H$ results in a positive semi-definite matrix. This indicates that the term $\mathrm{tr}(\mathrm{DGD}^H)$ in (40) is a convex quadratic function with respect to v.

To address the non-convexity of the unit modulus constraint (17d) in problem $P(3.1)$, it can be relaxed into a convex form, given by:

$$|v_n| \leq 1, \forall n = 1, 2, \ldots, N. \tag{41}$$

Therefore, problem $P(3.1)$ can be reformulated as:

$$P(3.2): \min_{v} v^H Z v - 2\,\mathrm{Re}\{v^H z\} - r, \tag{42a}$$

$$s.t. \quad \mathrm{tr}\left(\mathrm{AGA}^H\right) \geq \gamma_{r,th}\left(v^T\left((h_{r,b}^H h_{r,b}) \odot ((h_{b,r} G h_{b,r}^H)^T)\right) v^* + \sigma_r^2 N_r\right), \tag{42b}$$

$$|v_n| \leq 1, \quad \forall n = 1, 2, \ldots, N. \tag{42c}$$

It can be observed that Z is a positive semi-definite matrix. In each iteration, problem $P(3.2)$ is a convex problem, which can be directly solved using CVX. The obtained phase shift solution for the IRS might violate the unit modulus constraint upon convergence. To remedy this, the amplitude of each IRS phase shift is normalized to unity, represented as:

$$v_n^{\mathrm{opt}} = \frac{v_n}{|v_n|}, \forall n = 1, 2, \ldots, N. \tag{43}$$

After normalization, the reconstructed IRS phase-shift vector satisfies the hardware-feasible unit-modulus constraint. Since this projection may change the radar SINR, normalization alone does not guarantee sensing feasibility. When the alternating optimization proceeds to the next iteration, the normalized IRS phase-shift matrix is used as the input to the BS active beamforming update, where problem $P(2.1)$ is solved with the radar SINR constraint explicitly imposed. In this manner, the sensing requirement is imposed again in the subsequent alternating-optimization step rather than being assumed to be preserved by normalization itself. Accordingly, the present work does not claim that the radar SINR constraint is preserved immediately after projection; developing a feasibility-preserving IRS recovery method with such a guarantee is left for future work. The proposed AOSF algorithm alternately optimizes the BS active beamforming matrix and the IRS passive beamforming matrix until convergence. The specific details of the proposed algorithm are outlined in Algorithm 1.

Algorithm 1 The proposed AOSF algorithm based on SDR, SCA, and FP.

1: **While** $R^{(k+1)} - R^{(k)} \geq \mu$ and $k \leq K$ do.
2: **Phase 1:** BS Active Beamforming Optimization:
3: Relax the rank-1 constraint in problem P(2.1) to transform it into a convex problem.
4: **Phase 2:** IRS Passive Beamforming Optimization:
5: Compute the updated Lagrange dual transformation auxiliary variable $b_m^{(k)}$ using Eq. (31).
6: Compute the updated quadratic transformation auxiliary variable $c_m^{(k)}$ using Eq. (34).
7: Employ the CVX convex optimization solver to address the convex problem $P(3.2)$ and update the variable $\boldsymbol{\Theta}^{(k)}$.
8: Calculate achievable sum-rate $R^{(k)}$ using $W^{(k)}$ and $\boldsymbol{\Theta}^{(k)}$; k = k + 1.

Based on the two main update stages in Algorithm 1, the overall computational complexity of the proposed AOSF algorithm can be expressed as $\mathcal{O}\left(I_{\mathrm{AO}}\left(C_{\mathrm{BS}} + C_{\mathrm{IRS}}\right)\right)$, where I_{AO} denotes the number of outer alternating iterations, and C_{BS} and C_{IRS} denote the computational costs of the BS and IRS subproblems, respectively. After SDR, problem $P(2.1)$ contains $M + N_T$ positive semidefinite matrix variables of size $N_T \times N_T$ and $2M$ auxiliary scalar variables. Its approximate scalar variable dimension is therefore $(M + N_T)N_T^2 + 2M$. For a dense interior-point implementation with a fixed number of solver iterations, C_{BS} scales approximately as $\mathcal{O}\left(\left((M + N_T)N_T^2 + 2M\right)^3\right)$.

Problem $P(3.2)$ contains an N-dimensional complex IRS phase-shift vector, corresponding to $2N$ real variables. Under the same dense-solver assumption, C_{IRS} scales approximately as $\mathcal{O}\left((2N)^3\right)$. The unit-modulus normalization requires only element-wise vector operations and therefore does not change the dominant complexity order. Since the radar SINR constraint is enforced within problem $P(2.1)$, the corresponding computational cost is included in C_{BS}.

The responses of the L sensing targets are aggregated into the sensing matrix A. Thus, increasing L increases the cost of constructing the sensing matrix but does not change the dimensions of the two principal optimization variables. Consequently, the BS-side SDP becomes the main computational bottleneck as N_T and M increase, whereas the IRS-side computational cost is mainly affected by the number of reflecting elements N. The convergence results further show that the proposed algorithm terminates within a limited number of outer iterations for the considered system scale.

5. Performance evaluation

This section provides simulation results to demonstrate the effectiveness of the proposed joint beamforming strategy within an IRS-assisted vehicular ISAC framework.

The system parameters are as follows: the three-dimensional coordinates of the VUEs are randomly generated within a circular area with

Table 1
Simulation parameters.

Parameters	Values
Number of BS transmitting antennas: N_T	8
Number of BS receiving antennas: N_R	8
Number of VUE: M	4
Number of IRS reflection elements: N	20
Noise Power at VUE: σ_m^2	-80 dBm
Noise Power at the Radar: σ_r^2	-80 dBm
Rician Factor: ζ	10 dB
BS Transmit Power: $P_{\max}$	0 dBm
BS location	(0m, 50m, 3m)
IRS location	(60m, 10m, 6m)

a radius of 10 m, centered at $(d_x\,\mathrm{m}, 0\,\mathrm{m}, 1.5\,\mathrm{m})$. Following the parameters specified in [28], the radar cross section is assigned a value of 1, while the sensing performance threshold is configured as $\gamma_{r,th} = 5$ dB. The convergence accuracy of the alternating optimization algorithm is set to $\mu = 10^{-5}$. In the simulation scenario, there are $L = 3$ sensing targets, located at $\theta_1 = -45^\circ$, $\theta_2 = 0^\circ$, and $\theta_3 = 45^\circ$, respectively. Additionally, the Rician factor is set to $\zeta = 10$ dB, and the path loss coefficients in the model are denoted as $\kappa_0 = -30$dB. The path loss exponents are set as $\alpha_{b,r} = 2.2$ for the BS-IRS link, $\alpha_{r,m} = 2.8$ for the IRS-to-vehicle link, and $\alpha_{b,m} = 3.5$ for the BS-to-vehicle link, in accordance with [39]. The simulation parameter settings are detailed in Table 1.

The three-target setting with $L = 3$, $\theta_1 = -45°$, $\theta_2 = 0°$, and $\theta_3 = 45°$ is adopted as a representative multi-target sensing case for evaluating the beampattern and the communication-sensing tradeoff. In more dynamic or cluttered environments, additional targets and multipath components may require more sensing resources, thereby reducing the degrees of freedom available for communication. The proposed framework can be extended to other target configurations by updating the target angles and the corresponding radar-SINR-related sensing matrix, while clutter-aware and target-tracking-based robust beamforming will be considered in future work.

It should be noted that the BS transmit power $P_{\max} = 0$ dBm in Table 1 is used only as a reference operating point for the basic power-constrained setup. The evaluation is not limited to this value, since the impact of different BS transmit power levels is further investigated in Fig. 4, where the proposed AOSF algorithm consistently outperforms the ISAC benchmark schemes.

To comprehensively assess the proposed algorithm, four benchmark schemes are considered from the perspectives of the beamforming optimization strategy, IRS phase optimization, IRS deployment, and sensing constraints:

- **AO-ISAC:** This scheme is a conventional AO benchmark constructed under the system model and constraints considered in this paper. The BS active beamforming matrix W and the IRS passive beamforming matrix $\boldsymbol{\Theta}$ are alternately updated under the same BS transmit-power, radar-SINR, and IRS unit-modulus constraints. The relaxed IRS solution is recovered through unit-modulus projection. Compared with this conventional AO implementation, the proposed AOSF algorithm employs an FP-based passive beamforming update.
- **Random Phase Shift:** In this scheme, the phase shifts of the IRS are randomly generated within the range $[0, 2\pi)$ in accordance with a uniform distribution.
- **Without IRS:** This scheme includes scenarios where the IRS is not deployed (N = 0), and the vehicular ISAC system operates autonomously without IRS assistance.
- **IRS-Aided Only-Communication (com-only):** Here, the IRS is solely responsible for transmitting communication signals, without considering the constraints imposed by sensing performance. The optimization of the BS's active beamforming matrix W and the IRS's passive beamforming vector $\boldsymbol{\Theta}$ is performed iteratively via the Block Coordinate Descent (BCD) approach, as detailed in [40].

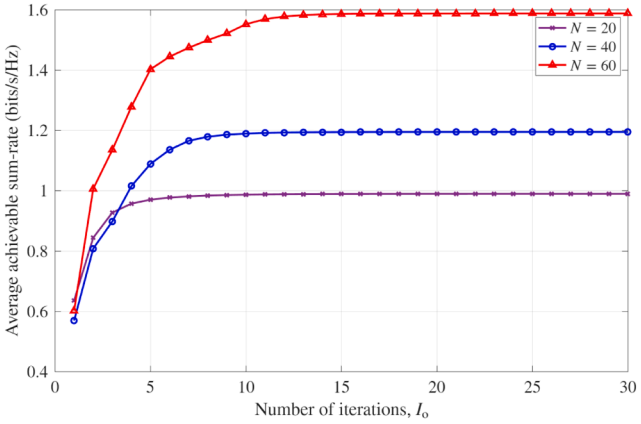


Fig. 2. The convergence process of the proposed algorithm under different numbers of IRS reflection elements.

In practical CAV scenarios, imperfect CSI may arise from channel estimation errors, vehicle mobility, residual Doppler effects, and IRS-assisted cascaded CSI acquisition. For passive IRS-assisted links, obtaining the BS-IRS-VUE cascaded CSI may introduce pilot overhead, especially when the number of IRS reflecting elements is large; this overhead can be reduced by IRS element grouping, location-aided channel estimation, beam training, or predicted CSI. CSI uncertainty may degrade the beam alignment accuracy of the BS active beamforming matrix W and the IRS passive beamforming matrix $\boldsymbol{\Theta}$, thereby reducing both achievable sum rate and sensing performance. Robust beamforming under bounded or statistical CSI errors will be considered in future work. In addition, the computational delay of the proposed AOSF algorithm should be evaluated with respect to the effective channel coherence time T_c^{eff}. Under the adopted block-fading model, the algorithm is executed for each quasi-static channel realization, and the computational time T_{comp} should be smaller than or comparable to T_c^{eff} for online operation. In high-mobility CAV scenarios, $\boldsymbol{\Theta}$ can be updated at a slower timescale, while W can be updated more frequently according to instantaneous or predicted CSI, which helps reduce signaling and computational overhead while adapting to vehicle mobility.

The convergence performance of the proposed algorithm for varying numbers of IRS reflecting elements with $d_x = 65$ m is shown in Fig. 2. The simulation results indicate that the proposed algorithm generally achieves convergence within 15 iterations across different IRS element configurations, confirming its reliable convergence behavior. As the number of IRS-reflecting elements increases, the achievable rate of the proposed algorithm also increases linearly. For example, at 20 iterations, the system's achievable rate increases by 37.7% for $N = 60$ compared to $N = 20$. This enhancement stems from the increased number of IRS reflecting elements, which leverage additional spatial degrees of freedom to capture and reflect more of the BS's received signal power, resulting in improved passive beamforming gains. A smaller number of IRS reflecting elements reduces the dimension of the passive beamforming subproblem and therefore lowers the computational cost per iteration.

The comparison of convergence performance between the proposed algorithm and benchmark algorithms is shown in Fig. 3 for $d_x = 65$ m. As shown in the figure, all schemes converge within 30 iterations. Compared with the AO-ISAC benchmark, the proposed AOSF algorithm converges to a higher achievable sum rate. This indicates that the proposed joint active and passive beamforming design can obtain better communication performance while explicitly accounting for the radar SINR requirement in the constrained beamforming subproblems. The proposed algorithm demonstrates a 30.3% improvement in the achievable rate compared to the no-IRS scheme. This improvement can be attributed to the additional BS-IRS-communication vehicle reflection links introduced by the IRS, which improve downlink communication performance. In

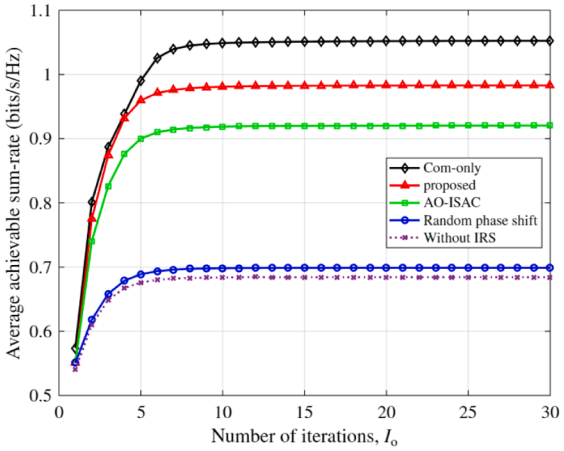


Fig. 3. The comparison of convergence performance of different algorithms.

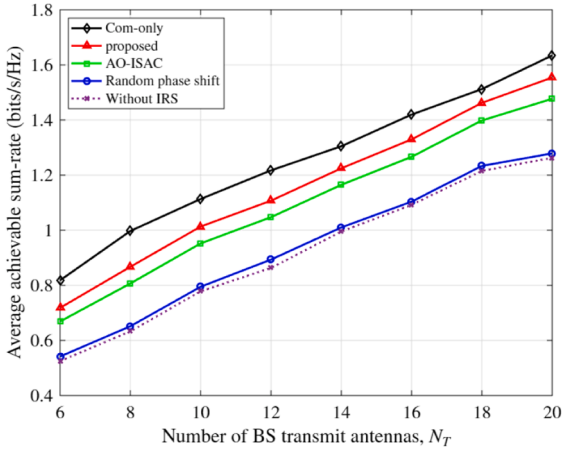


Fig. 5. The average achievable sum-rate versus the number of BS transmit antennas.

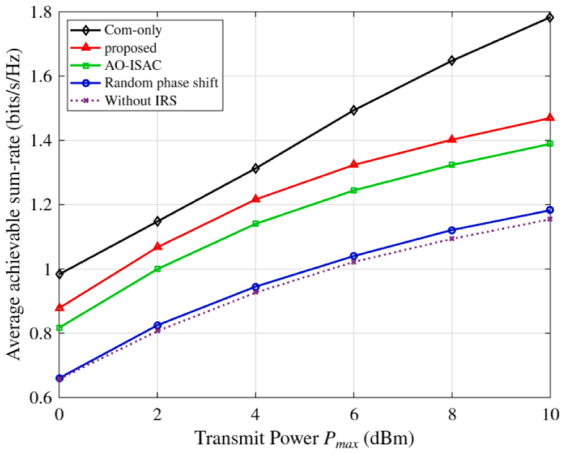


Fig. 4. The average achievable sum-rate versus the transmit power of BS.

contrast, the IRS random phase shift scheme yields only marginal improvement in performance compared to the no-IRS scheme. This occurs because, in the absence of passive beamforming at the IRS, the signals reaching the IRS cannot be accurately directed toward the VUEs, thereby underscoring the necessity of passive beamforming at the IRS. Moreover, because of the inherent trade-off between communication and radar sensing capabilities, a performance difference arises when comparing the considered ISAC system with an IRS-assisted communication-only system. This gap arises because the considered ISAC system, while maintaining the same BS transmit power, must allocate additional resources for radar sensing, which in turn reduces the achievable rate for VUEs.

The correlation between the system's achievable rate and the base station's transmit power for $d_x = 65$ m is shown in Fig. 4. It is evident that the achievable rate increases with higher transmit power levels. The proposed AOSF algorithm also outperforms the AO-ISAC benchmark over the whole transmit-power range, which demonstrates that the proposed algorithm can more effectively exploit the increased BS transmit power under the sensing constraint. This improvement can be attributed to higher power levels strengthening the received signal at the VUEs. In addition, the proposed algorithm demonstrates superior performance compared to both the random phase shift approach and the scenario without IRS assistance. For instance, at $P_{\max} = 10$ dBm, the proposed algorithm achieves an increase in achievable rate of 22.7% and 24.4% over the random phase shift scheme and the no-IRS scheme, respectively. This illustrates the effectiveness of the proposed algorithm in joint beamforming design for IRS-aided vehicular ISAC systems.

The relationship between the system's achievable rate and the number of BS transmit antennas for $d_x = 65$ m is shown in Fig. 5. It is evident that increasing the number of transmit antennas at the BS leads to higher achievable rates across all schemes. Compared with the AO-ISAC benchmark, the proposed AOSF algorithm achieves a higher achievable sum rate under different numbers of BS transmit antennas, showing that the proposed design can better utilize the additional spatial degrees of freedom provided by multiple BS antennas. To illustrate, when the number of BS transmit antennas $N_T = 12$, the achievable rate of the proposed algorithm increases by 21.6% compared to when $N_T = 8$. This improvement results from the greater spatial degrees of freedom offered by the increased number of transmit antennas at the BS, which boost the active beamforming gain. Additionally, the proposed algorithm reliably achieves better performance compared to both the random phase shift method and the scenario without IRS. For instance, with 16 BS transmit antennas, the achievable rate of the proposed algorithm is 17.0 and 17.9 percent higher than that of the random phase shift scheme and the no-IRS scheme, respectively. This demonstrates the significant advantages of integrating an IRS into vehicular ISAC systems. Moreover, in an IRS-assisted communication-only system, the BS utilizes its full transmit power for transmitting information to VUEs, with the additional BS transmit antennas dedicated solely to communication functions. Hence, the achievable rate of the IRS-assisted communication-only scheme serves as an upper performance limit for all the schemes under consideration.

The achievable rate per IRS reflecting element at $d_x = 65$ m is depicted in Fig. 6. It is evident that in the proposed algorithm, the achievable rate increases with the number of IRS reflecting elements. The proposed AOSF algorithm also achieves better performance than the AO-ISAC benchmark as the number of IRS reflecting elements increases. This indicates that the proposed passive beamforming optimization can more efficiently exploit the IRS reflection gain under the unit-modulus phase-shift constraint. This behavior results from the IRS utilizing extra spatial degrees of freedom to enhance the reflection of the BS's received signal power, leading to increased passive beamforming gains. For instance, when the number of IRS reflecting elements $N = 40$, the achievable rate of the proposed algorithm increases by 39.1% compared to the random phase shift scheme. This enhancement is due to the constrained performance of the random phase shift scheme, where the IRS's random phase adjustments fail to consistently match the channels of the VUEs.

The relationship between the achievable rate and the radar SINR threshold at $d_x = 65$ m is shown in Fig. 7. A clear tradeoff can be observed between communication efficiency and radar sensing capability. As the radar SINR threshold $\gamma_{r,th}$ increases, the sensing requirement becomes more stringent, and more spatial and power resources need to be

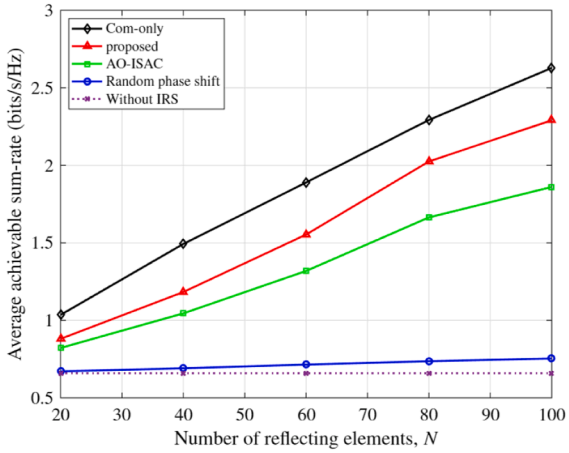


Fig. 6. The average achievable sum-rate versus the number of reflecting elements of IRS.

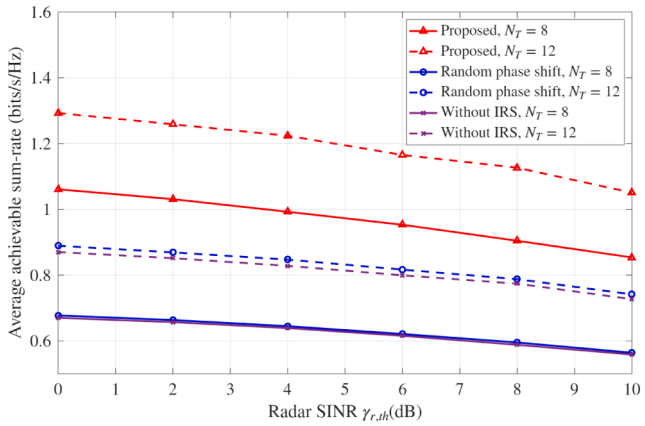


Fig. 7. The average achievable sum-rate versus radar SINR threshold.

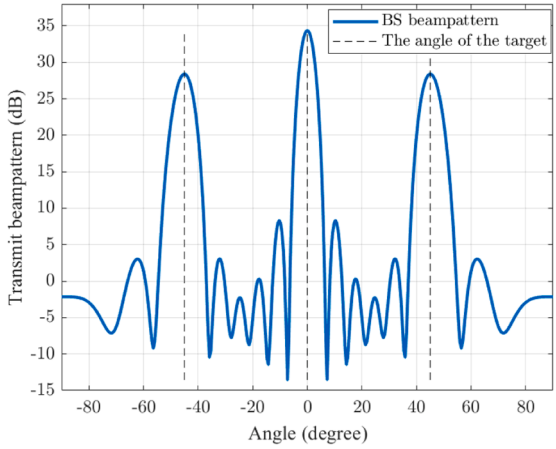


Fig. 8. Transmit beampattern.

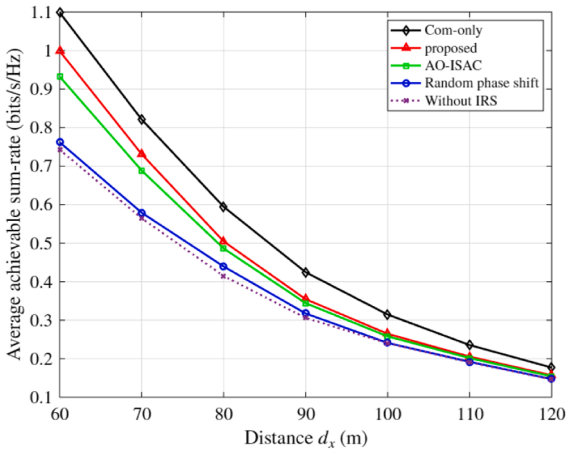


Fig. 9. The average achievable sum-rate versus the position of VUEs.

allocated to guarantee radar detection performance. Consequently, the achievable rate of the ISAC system decreases. For example, when $N_T = 8$ BS antennas, the achievable rate of the proposed algorithm decreases by 12.3% when $\gamma_{r,th} = 8$ dB compared with $\gamma_{r,th} = 2$ dB. This reduction is mainly because the joint beamforming design needs to reserve more resources for satisfying the radar SINR constraint, which reduces the degrees of freedom available for communication sum-rate maximization. Therefore, Fig. 7 not only evaluates the communication rate performance, but also verifies the impact of the sensing requirement on the proposed IRS-assisted ISAC design. Additionally, in scenarios where $N_T = 12$ BS transmit antennas, the system achieves better performance compared to the $N_T = 8$ scenario. This improvement can be attributed to the increased number of BS transmit antennas, which provide more spatial degrees of freedom and enhance beamforming gain.

To further validate the radar sensing performance, the beampattern is presented in Fig. 8. The BS transmit beampattern illustrates the power distribution of the transmitted signal over different angular directions and can be mathematically expressed as:

$$P_b(\theta_l; W) = \mathbb{E}\left\{\left|a_t^H(\theta_l) W s\right|^2\right\} = a_t^H(\theta_l) W W^H a_t(\theta_l). \tag{44}$$

As shown in Fig. 8, the BS maintains high power levels around the target directions by forming radar beams toward the sensing targets, while reducing the power radiated toward irrelevant directions. This result demonstrates that the proposed joint beamforming design can support effective radar sensing while serving VUEs, thereby further confirming the communication-sensing balance achieved by the proposed IRS-assisted ISAC framework.

The connection between the achievable rate and the locations of VUEs is illustrated in Fig. 9. The results indicate that for all schemes, the achievable rate declines as the distance between the VUEs and both the BS and IRS grows. Although the achievable rate decreases when the VUEs move farther away, the proposed AOSF algorithm consistently outperforms the AO-ISAC benchmark at different VUE positions. This confirms the effectiveness of the proposed joint beamforming design under different vehicular user locations. Specifically, when m increases, the achievable rate of the proposed algorithm drops by 30.9% compared to scenarios with smaller m. This decline is mainly due to the increased distance, leading to higher path losses in the signal transmission path. Consequently, the signal strength from the BS to the VUEs, as well as from the IRS to the VUEs, gradually diminishes, which results in a reduction in the achievable rate.

6. Conclusions

This work investigates the joint optimization of active and passive beamforming in IRS-assisted ISAC systems for CAVs. The objective is to determine the optimal solution that maximizes the achievable sum rate of VUEs, while satisfying the power budget at the BS, the radar's minimum SINR constraint, and the IRS unit-modulus phase-shift constraint. To achieve this, we propose the AOSF algorithm, which employs semidefinite relaxation and successive convex approximation algorithms to optimize the BS active beamforming matrix, and a fractional programming algorithm to optimize the IRS passive beamforming matrix. This approach transforms the non-convex problem into a series of manageable subproblems, which are then solved iteratively.

Simulation outcomes confirm the effectiveness of incorporating an IRS into CAV-ISAC systems. In contrast to benchmark schemes, including the AO-ISAC benchmark, the without-IRS scheme, and the random phase shift scheme, the proposed method achieves enhanced achievable sum-rate performance. This highlights the inherent trade-off between communication throughput and sensing accuracy. Future research will extend to vehicle target tracking, with the aim of incorporating parameters such as vehicle speed, position, and direction estimation into the realm of sensing tasks. This will provide more accurate sensing performance and achieve more precise beamforming prediction, enhancing the quality of the services provided by CAV communications.

CRediT authorship contribution statement

Yaping Cui: Writing – review & editing, Writing – original draft, Supervision; **Jianjun Gu:** Writing – original draft; **Dapeng Wu:** Validation, Resources, Investigation; **Peng He:** Supervision, Resources, Data curation, Conceptualization; **Ruyan Wang:** Validation, Investigation, Funding acquisition, Data curation; **Gongxun Wang:** Visualization, Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] C. Yang, et al., Navigating the road ahead: a comprehensive survey of radio resource allocation for vehicle platooning in C-V2X communications, IEEE Commun. Surv. Tutor. 27 (2) (2025) 1326–1362.
[2] C. He, et al., Joint design of communication efficiency and resource allocation in V2X networks based on multi-agent deep reinforcement learning, in: Proc. IWCMC, 2025, pp. 509–513.
[3] D.P.M. Osorio, et al., The rise of networked ISAC: emerging aspects and challenges, IEEE Open J. Commun. Soc. 6 (2025) 5072–5091.
[4] D. Cong, et al., Vehicular behavior-aware beamforming design for integrated sensing and communication systems, IEEE Trans. Intell. Transp. Syst. 24 (6) (2023) 5923–5935.
[5] B. Mendes, et al., Exploring V2X in 5G networks: a comprehensive survey of location-based services in hybrid scenarios, Veh. Commun. 52 (2025) 100878.
[6] A. Tang, et al., Interference management for full-duplex ISAC in B5G/6G networks, IEEE Commun. Mag. 62 (9) (2024) 20–26.
[7] G. Liu, et al., Sensing accuracy gain, unified performance analysis and optimization in 6G cooperative ISAC systems, Digit. Commun. Netw. 11 (5) (2025) 1657–1667.
[8] M.U.F. Qaisar, et al., ISAC-assisted wireless rechargeable sensor networks with multiple mobile charging vehicles, IEEE Internet Things Mag. 7 (6) (2024) 80–86.
[9] N.C. Luong, et al., Advanced learning algorithms for integrated sensing and communication (ISAC) systems in 6G and beyond: a comprehensive survey, IEEE Commun. Surveys Tutor. 28 (2026) 2572–2611.
[10] Y. Zhang, et al., Research and experimental validation for 3GPP ISAC channel modeling standardization, Digit. Commun. Netw. 11 (5) (2025) 1601–1613.
[11] A. Liu, et al., A survey on fundamental limits of integrated sensing and communication, IEEE Commun. Surv. Tutor. 24 (2) (2022) 994–1034.
[12] X. Cheng, et al., Integrated sensing and communications (ISAC) for vehicular communication networks (VCN), IEEE Internet Things J. 9 (23) (2022) 23441–23451.
[13] M.B. Mollah, et al., mmWave-enabled connected autonomous vehicles: a use case with V2V cooperative perception, IEEE Netw. 38 (6) (2024) 485–492.
[14] R. Zhang, et al., Generative AI-enabled vehicular networks: fundamentals, framework, and case study, IEEE Netw. 38 (4) (2024) 259–267.
[15] A. Zakerimanesh, et al., Rigid communication topologies: impact on stability, safety, energy consumption, passenger comfort, and robustness of vehicular platoons, Veh. Commun. 54 (2025) 100936.
[16] A. Al-Hilo, et al., Reconfigurable intelligent surface enabled vehicular communication: joint user scheduling and passive beamforming, IEEE Trans. Veh. Technol. 71 (3) (2022) 2333–2345.
[17] N.P. Fall, et al., Exploring the use of mobile IRS in a vehicular context, Veh. Commun. 55 (2025) 100965.
[18] Z. Kang, et al., Active-IRS-aided wireless communication: fundamentals, designs, and open issues, IEEE Wirel. Commun. 31 (3) (2024) 368–374.
[19] B. Zheng, et al., A survey on channel estimation and practical passive beamforming design for IRS aided communications, IEEE Commun. Surv. Tutor. 24 (2) (2022) 1035–1071.
[20] S. Gong, et al., Toward smart wireless communications via intelligent reflecting surfaces: a contemporary survey, IEEE Commun. Surv. Tutor. 22 (4) (2020) 2283–2314.
[21] Y. Chen, et al., Robust beamforming for active reconfigurable intelligent omni-surface in vehicular communications, IEEE J. Sel. Areas Commun. 40 (10) (2022) 3086–3103.
[22] R. Liu, et al., Integrated sensing and communication with reconfigurable intelligent surfaces, IEEE Wirel. Commun. 30 (1) (2023) 50–57.
[23] A.M. Elbir, et al., The rise of intelligent reflecting surfaces in integrated sensing and communications paradigms, IEEE Netw. 37 (6) (2022) 224–231.
[24] H. Luo, et al., Joint beamforming design for IRS-assisted integrated sensing and communication systems, IEEE Trans. Veh. Technol. 71 (12) (2022) 13393–13397.
[25] K. Zhong, et al., Joint waveform and beamforming design for IRS-aided ISAC systems, IEEE Signal Process. Lett. 30 (2023) 165–169.
[26] M. Hua, et al., Joint active and passive beamforming design for IRS-aided radar-communication, IEEE Trans. Wirel. Commun. 22 (4) (2022) 2278–2294.
[27] X. Song, et al., Joint transmit and reflective beamforming for IRS-assisted integrated sensing and communication, in: Proc. IEEE WCNC, 2022, pp. 189–194.
[28] H. Luo, et al., RIS-aided integrated sensing and communication: joint beamforming and reflection design, IEEE Trans. Veh. Technol. 72 (7) (2023) 9626–9630.
[29] K. Han, et al., Sensing-secure ISAC: ambiguity function engineering for impairing unauthorized sensing, IEEE Trans. Wirel. Commun. 25 (2026) 5386–5399.
[30] S. Abdallah, et al., Channel estimation for full-duplex OFDM-based ambient backscatter communication systems, IEEE Trans. Commun. 72 (9) (2024) 5302–5318.
[31] X. Liu, et al., Joint transmit beamforming for multiuser MIMO communications and MIMO radar, IEEE Trans. Signal Process. 68 (2020) 3929–3944.
[32] G. Zhou, et al., Channel estimation for IRS-aided multiuser millimeter-wave systems, IEEE Trans. Signal Process. 70 (2022) 1478–1492.
[33] S. Buzzi, et al., Foundations of MIMO radar detection aided by reconfigurable intelligent surfaces, IEEE Trans. Signal Process. 70 (2022) 1749–1763.
[34] L. Xu, et al., Target detection and parameter estimation for MIMO radar systems, IEEE Trans. Aerosp. Electron. Syst. 44 (3) (2008) 927–939.
[35] L. Zheng, et al., Joint design of overlaid communication systems and pulsed radars, IEEE Trans. Signal Process. 66 (1) (2017) 139–154.
[36] C. Jiang, et al., Secure beamforming design for IRS-assisted integrated sensing and communication systems, IEEE Wirel. Commun. Lett. 13 (2) (2023) 520–524. https://doi.org/10.1109/LWC.2023.3235880
[37] N.D. Sidiropoulos, et al., Transmit beamforming for physical-layer multicasting, IEEE Trans. Signal Process. 54 (6) (2006) 2239–2251.
[38] K. Shen, W. Yu, Fractional programming for communication systems—Part I: power control and beamforming, IEEE Trans. Signal Process. 66 (10) (2018) 2616–2630.
[39] Q. Wu, R. Zhang, Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming, IEEE Trans. Wirel. Commun. 18 (11) (2019) 5394–5409.
[40] H. Guo, et al., Weighted sum-rate maximization for reconfigurable intelligent surface aided wireless networks, IEEE Trans. Wirel. Commun. 19 (5) (2020) 3064–3076.



Yaping Cui received the M.S. degree in communication and information system and the Ph.D. degree in traffic information engineering and control from Southwest Jiaotong University, Chengdu, China, in 2011 and 2017, respectively. From 2011 to 2012, he was a Baseband Algorithm Engineer with ZTE Corporation, Shenzhen, China. In 2017, he joined the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, China, where he is currently an Associate Professor. His research interests include smart communications, mobile edge computing and network function virtualization for Connected and Autonomous Vehicles and NTN networks.



Jianjun Gu is currently pursuing the M.S. degree with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, China. His research interests include vehicular networks and Quality of Service Guarantee.



Dapeng Wu (Senior Member, IEEE) received the M.S. degree in communication and information systems from the Chongqing University of Posts and Telecommunications in June 2006, and the Ph.D. degree from the Beijing University of Posts and Telecommunications in 2009. His research interests include ubiquitous networks, IP QoS architecture, network reliability, and performance evaluation in communication systems.



Peng He received the Ph.D. degree from the University of Electronic Science and Technology of China, in 2018. He is currently a Lecturer with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications. His research interests include mobile edge computing, molecular communication, wireless body area networks. He is a member of IEEE ComSoc.



Ruyan Wang received the Ph.D. degree from the University of Electronic and Science Technology of China (UESTC) in 2007, and the M.S. degree from the Chongqing University of Posts and Telecommunications (CQUPT), China, in 1997. Since December 2002, he has been a Professor with the Special Research Centre for Optical Internet and Wireless Information Networks, CQUPT. His research interests include network performance analysis and multimedia information processing.



Gongxun Wang received the M.S. degree from the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, China, in 2024. He is currently working in Huangshi.